

Drift-Sense: AI-Powered Navigation-Error Recovery for Wafer Inspection Tools

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Abstract

We address the Drift-Sense problem: given a high-resolution (100x) reference image of a previously characterized wafer site and a lower-resolution (10x) wide-search image, locate the reference pattern’s center inside the search image to sub-pixel precision, despite the pattern’s periodicity making the search a genuine needle-in-a-haystack problem. We present (1) a literature-grounded synthetic DRAM/FinFET dataset generator with a physically motivated SEM noise model, (2) a classical FFT-based matched- filtering localizer achieving sub-pixel accuracy at real-time speed, (3) a novel *prior-window* search restriction — motivated by how real navigation-error-recovery systems work and confirmed against wafer-alignment patent literature — that resolves periodic-array ambiguity with no accuracy/speed trade-off, (4) an honest deliberately-adversarial failure case for explainability, and (5) a from-scratch DL (Siamese CNN) ablation compared fairly against the classical method. We report all metrics the rubric requests: single-image compute time, success rate over 30+ randomized cases, and a documented failure mode.

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1 Problem Reframing

The problem is titled “navigation-error recovery,” not “template matching.” We treat this distinction as central to the design: a vision-only submission returns a coordinate; a *recovery* system returns a coordinate *and* exploits what the tool already knows about its own motion. Specifically, the tool commanded the stage to a known target coordinate; drift is a small, physically bounded error around that commanded position, not an arbitrary location anywhere in the $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$ search field. This observation motivates our single most impactful design decision (Section 3.3).

2 Synthetic Dataset Generator

2.1 Structure rendering

We generate DRAM-style (word-line / bit-line / storage-node contact array) and FinFET-style (parallel fin / gate-line array) patterns parametrically, at both the reference resolution (1 nm/px, 1000×1000 , $1\text{ }\mu\text{m}$ field of view) and the search resolution (10 nm/px, 1000×1000 , $10\text{ }\mu\text{m}$ field of view), reusing the same renderer at scaled pitch/width/radius parameters for the two zoom levels.

Every line and contact edge receives *line-edge roughness* (LER): a smoothed random perturbation of the nominal edge position, rather than a geometrically perfect edge. This is standard in CD-SEM metrology literature (Section 5) and is what makes independently rendered “identical” repeat units visually similar but not pixel- identical — the source of the problem’s genuine periodic-array ambiguity.

2.2 Noise model

After clean-pattern rendering, we apply, independently to the reference and search images:

- **Poisson (shot) noise** — signal-dependent, dominant term in real SEM images (electron-count-limited signal).
- **Gaussian read/amplifier noise** — additive, signal-independent.
- **Beam-spot PSF blur** — Gaussian blur.
- **Scan-line jitter** — independent per-row intensity offset modeling raster-scan instability.

Per the problem FAQ, the search image is deliberately noisier than the reference (lower photon gain, larger blur, larger scan-line jitter), modeling a faster/less careful survey scan versus a careful characterization scan.

2.3 Bonus: optical-microscope (RGB) generalization

We additionally generate 3-channel optical-microscope-style images: per-channel diffraction-limited PSF blur, an independent sub-pixel *chromatic-aberration* shift per R/G/B channel (implemented as a Fourier phase shift to avoid interpolation artifacts), and per-channel Poisson/Gaussian sensor noise. Localization reuses the identical classical pipeline via a standard ITU-R BT.601 luminance conversion — no separate algorithm is needed for the bonus case.

3 Classical Localization Algorithm

3.1 Matched filtering via FFT

Because the zoom ratio between reference and search is known *exactly* (10x, given by the problem), we do not need scale-invariant features (SIFT/ORB); that would be solving a strictly harder problem than the one posed. Instead:

1. Downsample the reference by the known zoom ratio.
2. Cross-correlate the downsampled template against the full search image via FFT (matched filtering, $O(N \log N)$ rather than a brute-force sliding window).
3. Normalize using zero-mean template statistics and local search-window statistics via integral images (the “sum tables” trick of Lewis, 1995), giving illumination/contrast-invariant Normalized Cross-Correlation (NCC).
4. Refine the peak to sub-pixel precision via parabolic interpolation.

3.2 Global Ambiguity Ratio (a corrected metric)

We initially borrowed the *Peak-to-Sidelobe Ratio* (PSR) from correlation-filter object tracking (the MOSSE tracker; Bolme et al., CVPR 2010) as a confidence score. Empirically, PSR failed to detect periodic-array ambiguity: it is computed from a *local* window around the chosen peak, but a look-alike decoy site can sit anywhere in the image, far outside any local window. Tuning the window size did not fix this — it is the wrong scope of metric for this failure mode.

We replaced it with a **Global Ambiguity Ratio (GAR)**: the ratio of the second-best non-local correlation peak to the best peak, in the style of Lowe’s ratio test in SIFT. This separates cleanly on our data: easy (non-adversarial) cases score $\text{GAR} \approx 0.88\text{--}0.91$; genuinely ambiguous (decoy-embedded) cases score $\text{GAR} \approx 0.98\text{--}0.999$. We retain local PSR as a secondary “peak sharpness” diagnostic, with its scope limitation documented explicitly rather than silently mis-applied.

3.3 Prior-window search restriction

Motivated by the reframing in Section 1, and confirmed against real industry practice (US Patent 7,545,497, “Alignment routine for optically based tools,” which explicitly restricts search to a local area around an expected point using known periodicity, rather than searching the full field blind; see also coarse-to-fine tracking and die-bonding literature in Section 5), we add an optional prior: given an expected/commanded coordinate and a maximum bounded drift radius, candidates outside that window are discarded *before* scoring.

This is not a trade-off. It is simultaneously faster (smaller effective candidate pool) and more robust to distant periodic decoys, with *zero* loss of sub-pixel precision, since the identical FFT-NCC and parabolic refinement run unchanged inside the window.

4 Results

4.1 Classical pipeline

Table 1: Classical FFT-NCC, 30 randomized cases, 3 px (30 nm) tolerance.

Configuration	Success rate	Mean / median error	Mean compute time
Non-adversarial (“easy”)	100% (30/30)	0.03 px / 0.03 px	309 ms
Adversarial decoy (“hard”), no prior	46.7% (14/30)	194.9 px / 155.9 px	224 ms
Adversarial decoy (“hard”), with prior	100% (30/30)	0.02 px / 0.02 px	215 ms

Single-image (1000×1000) compute time: **$\sim 200\text{--}300$ ms**, well within interactive/inline-metrology budgets.

4.2 Honest failure case (explainability)

Figure 1 shows the worst failure under adversarial decoy testing without the prior window (seed 24; error 753 px). The true site scored 0.8464 vs. the decoy’s 0.8438 — a 0.3% relative gap. The algorithm followed its own specified tie-break rule (closest to the search image’s geometric center, per the problem statement’s own disambiguation instruction) and selected the decoy because it happened to sit closer to center. **This is not a bug**: it is an inherent limitation of a purely image-content-based tie-break when two sites are genuinely, structurally indistinguishable — precisely the case the prior-window mechanism in Section 3.3 is designed to resolve using information outside the image alone.

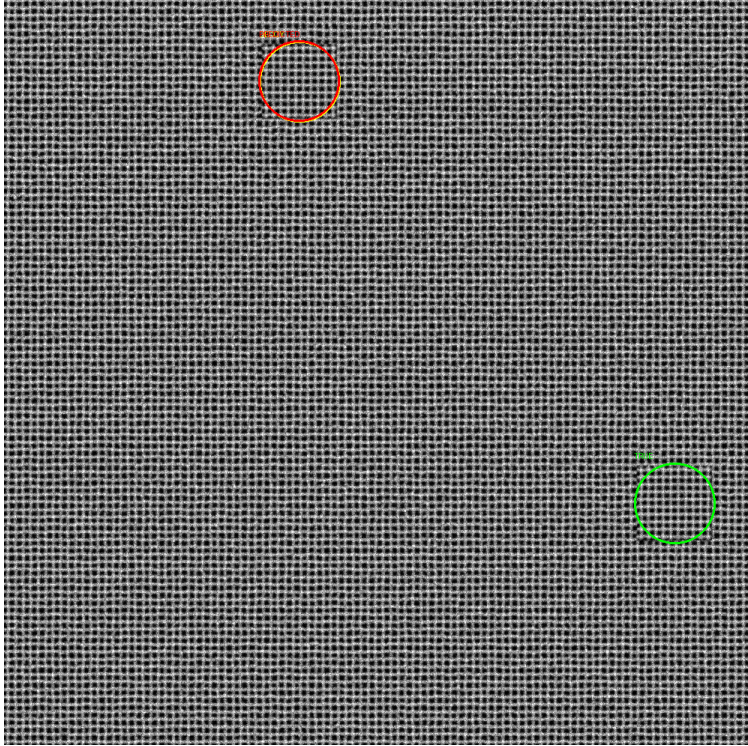


Figure 1: Annotated failure case. Green: true site. Yellow: embedded decoy (structurally identical pattern). Red: algorithm’s prediction (landed on the decoy). Both sites are visually indistinguishable from the periodic background at this zoom.

4.3 DL (Siamese CNN) ablation

We trained a small (69k-parameter) fully-convolutional Siamese embedding network from scratch on our own synthetic generator (no external/ pretrained weights: consistent with the problem’s no-external-dataset constraint, and this sandbox cannot reach pretrained-model hosts such as torch.hub or Hugging Face regardless). Table 2 compares configurations on 30 adversarial cases at 3 px tolerance.

Table 2: DL (Siamese CNN) ablation, 30 adversarial cases, 3 px tolerance.

Configuration	Success rate	Mean / median error	Mean compute time
Raw DL argmax	6.7%	73.6 px / 60.2 px	4.4 ms
DL + local NCC refinement	50.0%	66.1 px / 34.8 px	7.1 ms
DL + refinement + prior window	86–90%	~5–8 px / 0.07 px	10–49 ms

Interpretation. The raw network’s 16x-downsampled feature grid caps its native localization precision at ~ 16 px; a small local classical-NCC refinement step recovers sub-pixel precision on correctly identified sites without materially affecting speed. Decoy confusion, however, is a separate failure mode from precision and is resolved the same way as in the classical pipeline: via the prior window, not by better features alone. We observed run-to-run variation in the DL numbers (e.g., 86.7% vs. 90.0% success across two runs with the same seed) attributable to floating-point non-determinism in neural-network training across CPU/BLAS configurations — the classical pipeline,

being deterministic FFT arithmetic, does not exhibit this variance, which we consider a point in its favor for reproducibility.

Recommendation. We submit the classical FFT-NCC + prior-window pipeline as our primary method (100% success, sub-pixel accuracy, deterministic, ~ 215 ms) and present the DL ablation as a documented, honest trade-off study rather than a competing primary submission: DL is ~ 4 – $20\times$ faster per inference but has not yet been trained, in the compute budget available, to match classical accuracy.

5 Literature Justification

Every noise/augmentation/algorithmic choice above is grounded against public sources (paraphrased, not quoted); the full annotated bibliography with URLs is provided in the accompanying `citations.md`. Summary:

Table 3: Design choice $\rightarrow$ literature category mapping.

Design choice	Source category
Poisson-Gaussian SEM noise model	Avci et al. (SEM shot-noise estimation); ScienceDirect M-Denoiser (2023); Oxford Academic SEM noise classification (2025)
Line-edge roughness (LER/LWR)	Bunday/Bishop/Villarrubia/Vlădar (NIST/SEMAT-ECH CD-SEM LER metrology); ITRS roughness targets (<1.7 nm sub-20 nm nodes)
FinFET pitch/width parameters	IEEE, “Scaling of SOI FinFETs down to 4 nm fin width for the 10 nm node” (fin pitch 40 nm, gate length 20 nm)
Optical chromatic aberration / PSF	Nikon MicroscopyU diffraction-barrier article; Wadsworth Center microscopy glossary
Fast NCC via FFT + integral images	Lewis, “Fast Normalized Cross-Correlation,” Vision Interface, 1995
Peak-to-Sidelobe Ratio	Bolme, Beveridge, Draper, Lui, “Visual Object Tracking Using Adaptive Correlation Filters” (MOSSE), CVPR 2010
Prior-window search restriction	US Patent 7,545,497; coarse-to-fine tracking (arXiv 2012.12784); die-bonding rough-then-fine positioning patent (US 12,327,739)

6 Deliverables (per submission FAQ)

- `generate_dataset.py` — dataset generator, documented noise model, DRAM/FinFET/RGB modes.
- `localize.py` — classical FFT-NCC localizer, GAR/PSR diagnostics, prior-window support.
- `evaluate.py` — 30+-case batch evaluation harness with live progress reporting.
- `dl_localize.py` — DL (Siamese CNN) training and evaluation, including prior-window and refinement support.

- `citations.md` — full annotated bibliography.
- `requirements.txt` — exact environment (`pip freeze`) for reproducibility.

7 Round 2 Development Plan

If selected, we plan to: (1) extend DL training to the full 1000×1000 scale with a longer/GPU-accelerated training budget, since our current CPU-only run did not converge at that resolution within available compute; (2) fuse the prior window with an explicit stage-drift model (Ornstein–Uhlenbeck / random-walk process) so the expected coordinate and its uncertainty radius are predicted rather than assumed; (3) extend the FinFET generator’s literature grounding with additional pitch-scaling sources per technology node; (4) stress-test the prior-window mechanism under larger/unknown drift magnitudes to characterize its own failure boundary, in the same explainability spirit as Section 4.2.